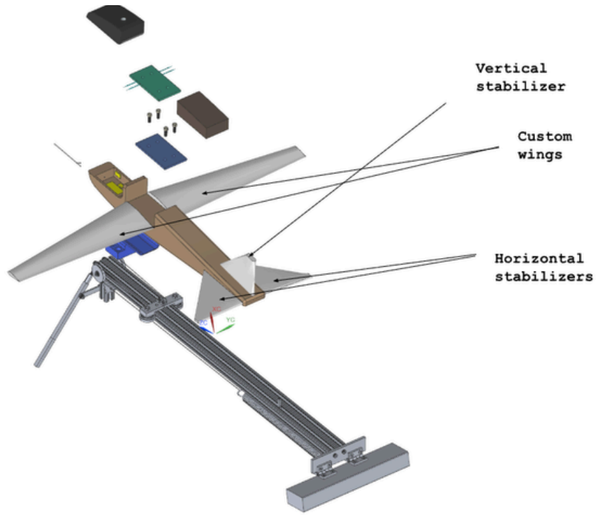
Maura Madden

Engineering & Design Portfolio





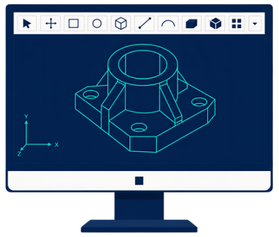




Mechanical Engineering · Product Design · CAD & Manufacturing · Prototyping · Human-Centred Design

London, UK | Email | LinkedIn

Skills Snapshot



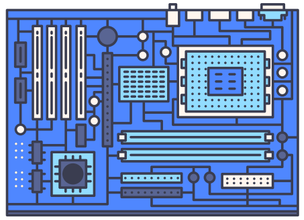
CAD & Design

SOLIDWORKS, Siemens NX, GD&T tolerancing, mould design



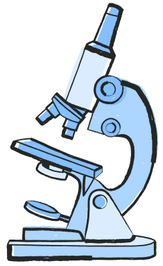
Manufacturing

CNC machining, CAM toolpaths, 3D printing, injection molding, cleanroom processing



Electronics

ESP32, Arduino, soldering, breadboarding, sensor integration, calibration, debugging



Testing & Research

Python, MATLAB, market research, user interviews, experimental testing,



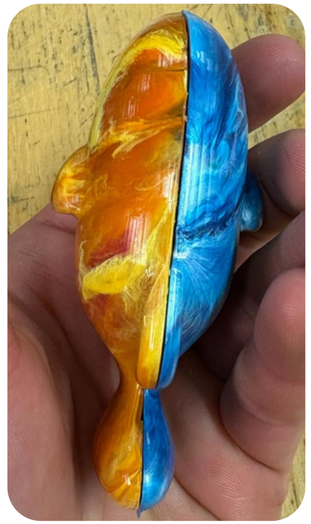
Design & Innovation

Human-centred design, concept generation, rapid prototyping, iterative testing, design evaluation

CNC Fish & Mould Design







Computer Integrated Manufacturing Course | Northwestern University

CAD/CAM & Manufacturing Engineer | January - April 2025

Project Overview

Designed and manufactured an interlocking, two-part fish assembly, taking the project from CAD modelling and mould design through CNC machining, injection moulding and dimensional inspection.

Key Outcomes

- 40 fish produced through injection molding
- 4 mould halves designed and CNC-machined
- 100% of measured features within specified tolerance limits

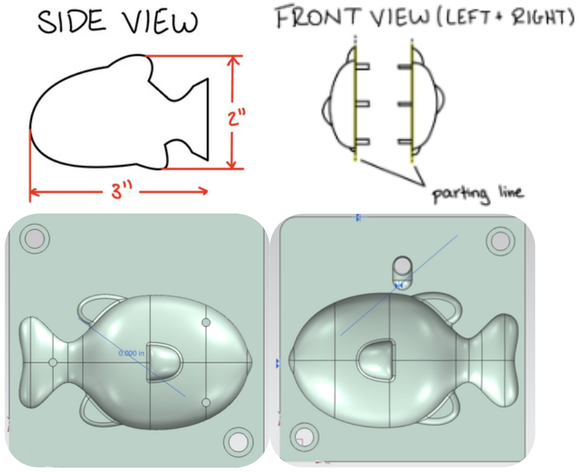
Skills Demonstrated

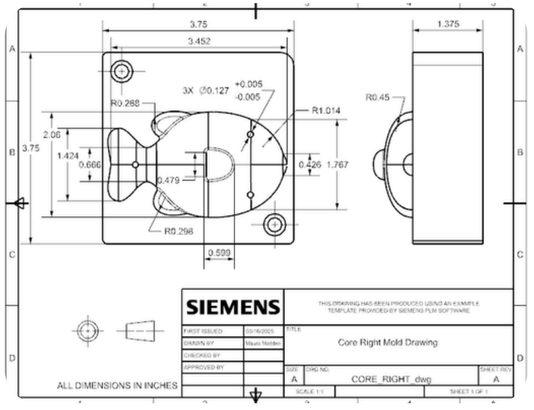
Siemens NX · CAD/CAM · Mould Design · CNC Machining · Injection Moulding · Dimensional Inspection · GD&T & Tolerancing

CNC Fish & Mould Design Process

1) CAD & Mould Design

Designed a two-part, interlocking fish assembly in Siemens NX with an interference fit, alongside four mould halves for injection moulding.

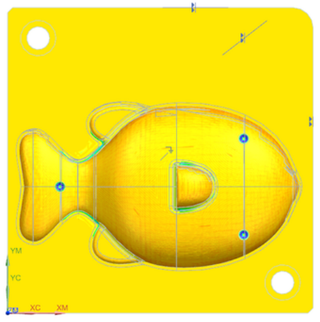


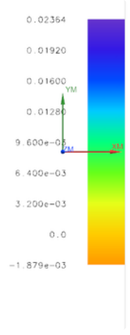


2) CAM & CNC Machining

Generated CNC toolpaths in Siemens NX and machined the moulds. Refined machining operations to improve dimensional accuracy and surface finish.

Title	Path	Tool	Description	Depth of Cut	Stepover	Feed	Speed
GEOMETRY			mill_contour				
Unused Items			mill_contour				
MCS_MILL			MCS_MILL				
WORKPIECE			WORKPIECE				
ROUGHING_1	✓	EM_0.75	CAVITY_MILL	.2 in	33 %Flat	18 ipm	2037 rpm
FINISHING_1	✓	EM_0.75	CAVITY_MILL	.2 in	33 %Flat	18 ipm	2037 rpm
SPOT_DRILLING	✓	EM_0.125	SPOT_DRILLING			10 ipm	1000 rpm
DRILLING	✓	D_0.125	DRILLING			4 ipm	2000 rpm
ROUGHING_2	✓	EM_0.5	CAVITY_MILL	.15 in	33 %Flat	16 ipm	3056 rpm
FINISHING_2	✓	EM_0.5	CAVITY_MILL	.15 in	33 %Flat	16 ipm	3056 rpm
ROUGHING_3	✓	BM_0.25	CAVITY_MILL	.07 in	33 %Flat	12 ipm	6112 rpm
AREA_MILL_1	✓	BM_0.25	AREA_MILL	10 %Tool	10 %Tool	12 ipm	6112 rpm
ROUGHING_4	✓	BM_0.125	CAVITY_MILL	.01 Scallop	33 %Flat	6 ipm	10000 rpm
AREA_MILL_2	✓	BM_0.125	AREA_MILL	10 %Tool	.01 Scallop	6 ipm	10000 rpm



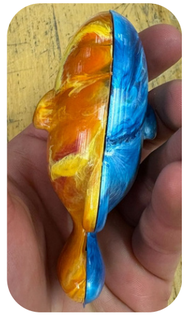


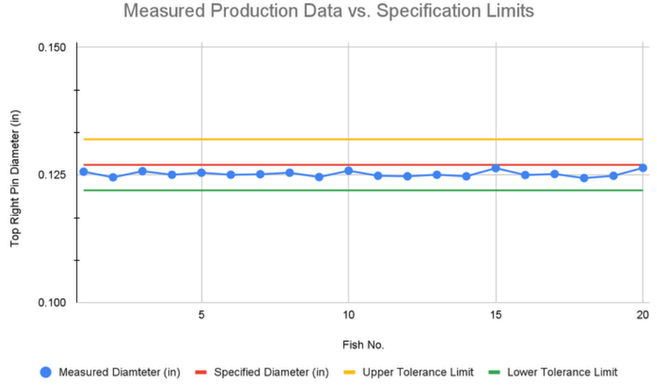


3) Injection Molding & Validation

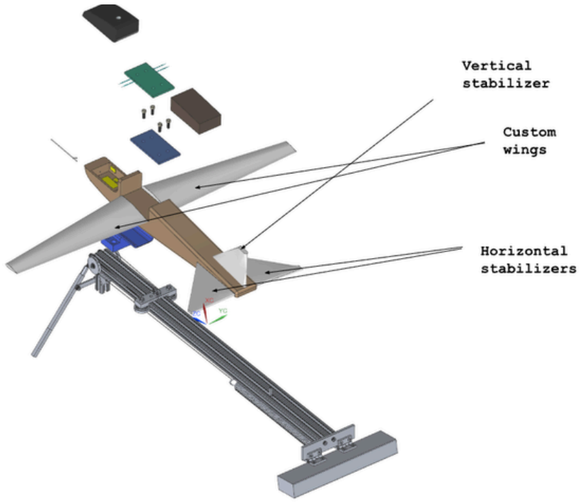
Produced 40 injection-moulded components and optimised processing parameters. Measured 40 moulded features across two component types, confirming that all measurements were within the specified tolerance limits.







Aerospace Testbed



Mechanical Engineering Capstone Project | Northwestern University

Systems Integration & Test Engineer | January – June 2025

Project Overview

Developed an aerospace testbed enabling high-school students to manufacture custom wings, conduct repeatable flight tests and investigate how wing design influences aircraft performance using sensor data.

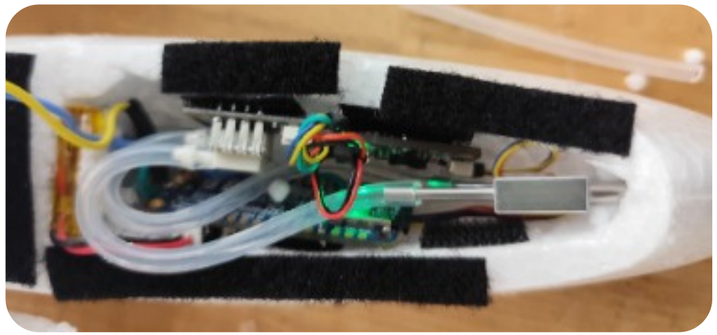
Our Three Integrated Subsytems:

1) Launching Mechanism



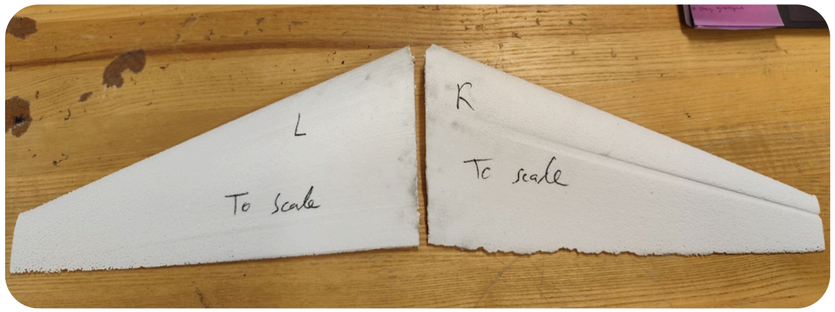
Developed a spring-powered launcher and carrier for repeatable aircraft launches.

2) Data Collection



Integrated onboard sensors and an ESP32 to capture flight data, and developed a web app to visualise results and analyse wing performance.

3) Wing Manufacturing Process



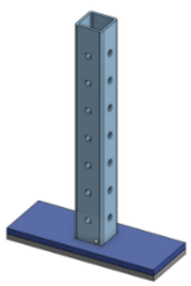
Evaluated fabrication methods and developed 2 jigs for producing custom tapered and non - tapered wing profiles.

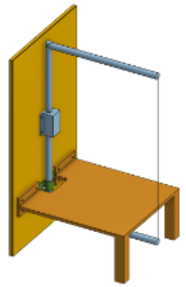
Aerospace Testbed Engineering Design & Validation

1) Wing Manufacturing & Testing

Compared 3D printing, laser-cut ribs and hot-wire foam cutting, assessing dimensional accuracy, cost, manufacturing time and weight. Designed and fabricated two jigs to support wing production.

Metric	Weight	3D Print		Foam	
		raw	weighted	raw	weighted
Cost per wing	0.1	5	0.5	5	0.5
Wing Dimensions Accuracy*	0.25	5	1.25	4	1
Wing Dimensions repeatability*	0.15	5	0.75	4	0.6
Wing manufacturing time	0.3	2	0.6	5	1.5
Weight	0.1	3	0.3	5	0.5
Airfoil Complexity	0.05	5	0.25	3	0.15
Strength	0.05	5	0.25	3	0.15
Sum	1		3.9		4.4
Rank			2		1







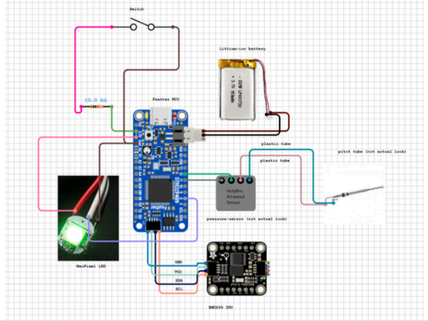


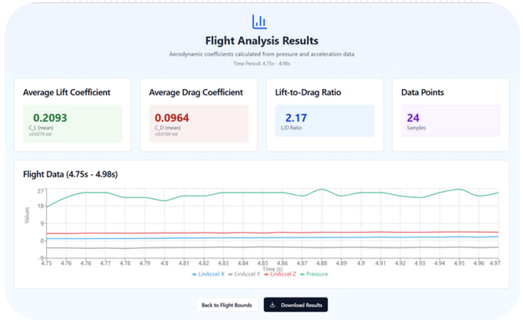
2) Electronics & Flight Data Analysis

Integrated an ESP32, IMU and pitot-tube pressure sensor to record flight data. Developed a web app for uploading and visualising flight data, selecting flight intervals and calculating lift and drag coefficients to compare wing performance.

Data Processing MCU Comparison

Solution	V_max	I_max	P_max	Weight	W	L	H	FLASH	CLK
Adafruit METRO 328	5V	.8A	4W	19g	2.1"	2.8"	.5"	32KB	16MHz
Arduino Nano	5V	.019A	.095W	7g	.7"	1.8"	.3"	32KB	16MHz
Adafruit ESP 32 Feather V2	5V	.25A	1.25W	6g	2.1"	.9"	.3"	8MB	240MHz
Raspberry Pi Pico 2W	5V	0.3A	1.5W	4g	2"	.83"	.3"	4MB	150MHz

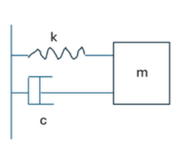




3) Launcher Design & Integration

Developed and tested a spring-powered launcher and aircraft carrier, integrating the launch mechanism with the selected fuselage for repeatable flight testing.

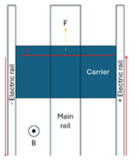
We identified 3 main approaches



$\frac{1}{2}kx^2 = E_{total}$

$x = 0.65\ m, k = 87\ N/m$

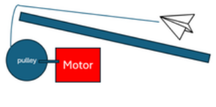
Feasible



$E_{total} \rightarrow F_{ave} * d_{launcher}$

$F_{ave} = ILB \rightarrow I \approx 350\ amps$

Not Feasible



$v_0 = \omega r_{pulley}$

$F_{ave} = \frac{\tau}{r_{pulley}}$

$\omega \approx 1600\ rpm, \tau \approx 1.5\ Nm$

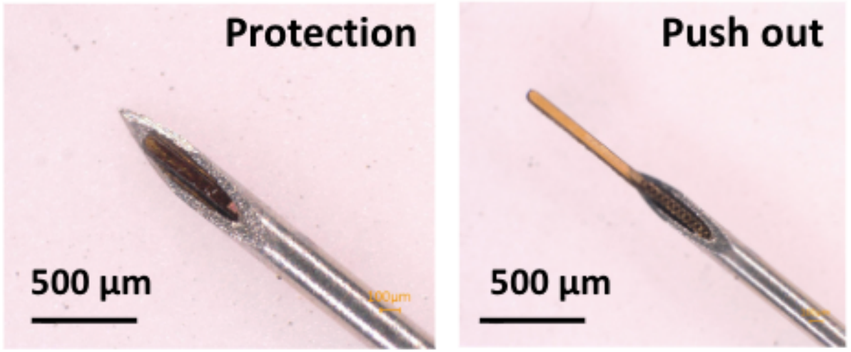
Feasible



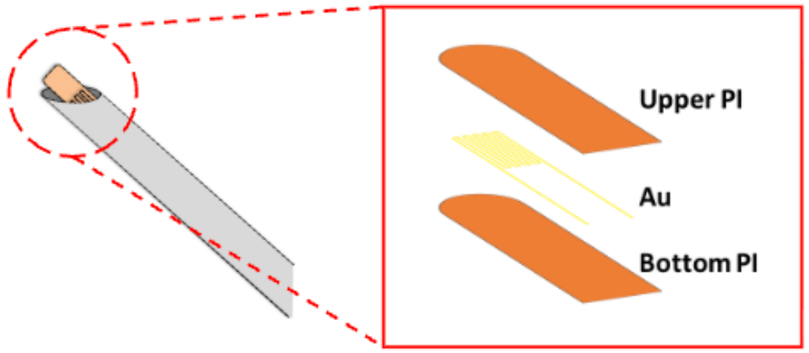


Minimally Invasive Blood Flow Sensor

Device image



Device Schematic



Querrey Simspon Institute for Bioelectronics | Northwestern University

Undergraduate Researcher at Rogers Research Group | January - September 2024

Project Overview

Contributed to the fabrication and benchtop testing of a minimally invasive piezoresistive sensor designed to monitor aortic blood flow during open-heart surgery.

Research Objective

Develop a proof-of-concept sensor capable of detecting changes in blood flow through the deflection of a microfabricated cantilever and corresponding changes in electrical resistance.

Sensor Working Princple

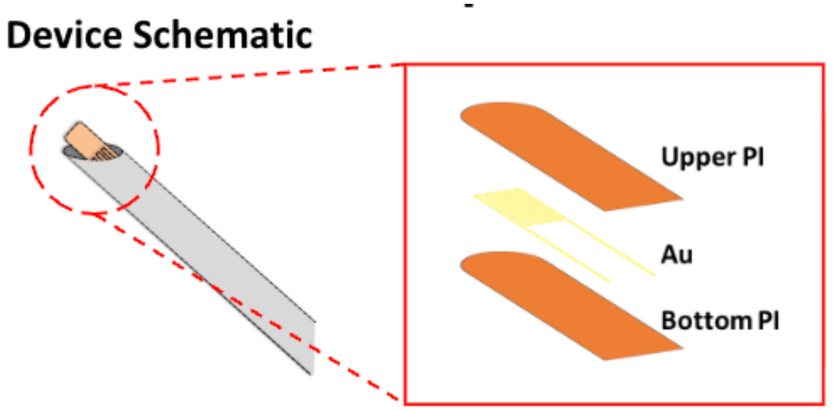
The project focuses on the mechanics of a piezoresistive sensor designed to measure blood flow dynamics. The sensor uses a cantilever mechanism with a gold sensing element, where flow-induced deflection changes the length and resistance of the metal strip, allowing blood flow direction and velocity to be inferred from electrical signals.

My Contribution

- Fabricated sensor devices using cleanroom microfabrication techniques .
- Supported benchtop testing using an artificial circulation system to assess sensor response under continuous and pulsatile flow conditions.

Fabrication & Experimental Validation

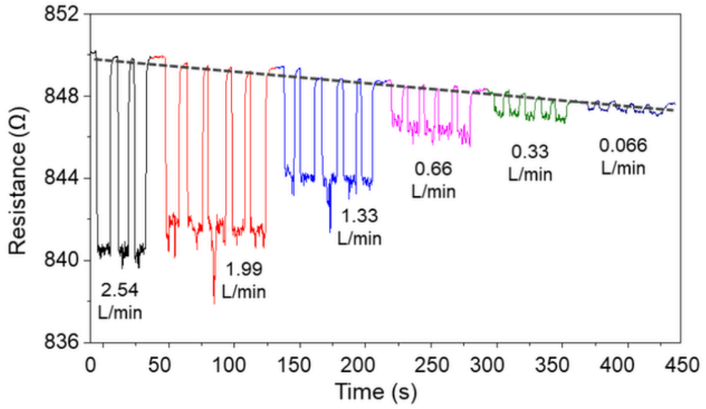
Device Schematic



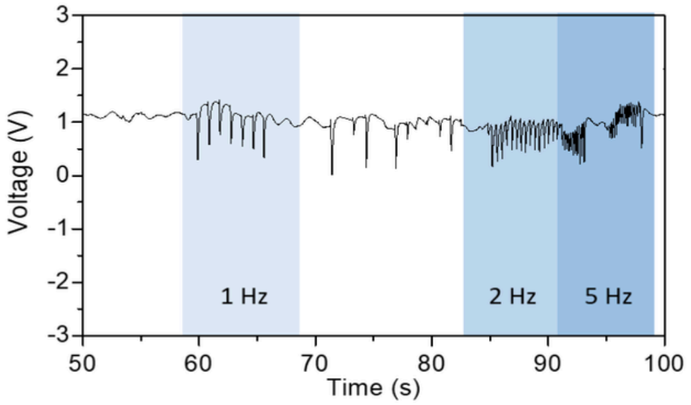
Device Fabrication

1. **Substrate** - PDMS, spin coating, and curing
2. **Photolithography** - photoresist coating, UV exposure and development
3. **Etching** - pattern transfer to form the cantilever
4. **Gold Deposition** - metal patterning and lift off
5. **Encapsulation** - Polyimide (PI) protective layer
6. **Wiring** - connection to readout circuitry
7. **Benchtop Testing** - sensor response under continous and pulsatile flow using an artificial circulation system.

Continuous Flow



Pulse Flow



Key Findings

Continuous flow

Measurable flow range of 0.066–2.54 L/min.

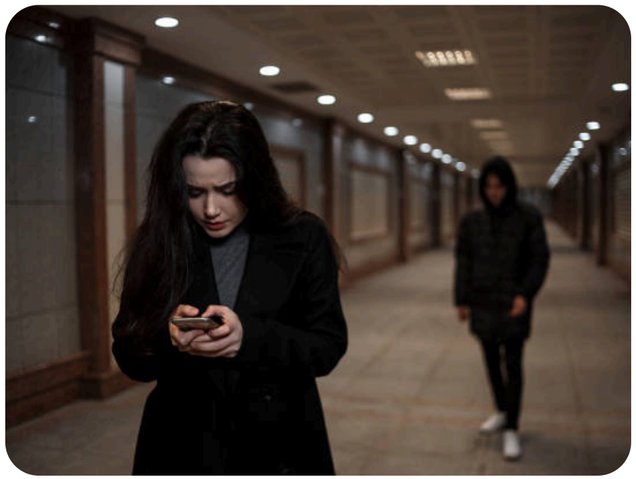
Pulsatile flow

Detected flow changes at frequencies up to 5 Hz during preliminary manual testing.

Limitation

Maximum continuous flow was limited by pump capacity.

AWear



Design Thinking & Communication Course | Northwestern University

Project Manager | April - June 2022

Design Challenge

Alert the user of potential danger when someone is behind them.

User Research

Conducted online research and interviewed women concerned about personal safety, alongside a police officer, to identify user needs and inform the design.

Key Insights

- Many women have reported that they do not feel safe while walking alone at night or running alone
- Police officer input informed a target detection distance of 15 ft.

Design Requirements

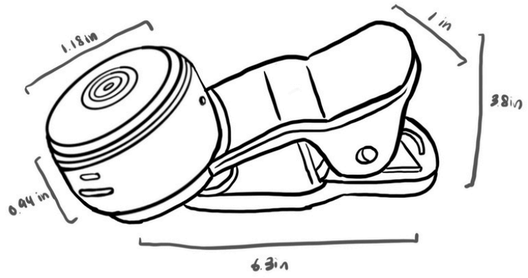
- 15 ft detection range
- Portable and discreet
- Notifies the user
- Easy to use

AWear

Clip on Motion Sensing Camera

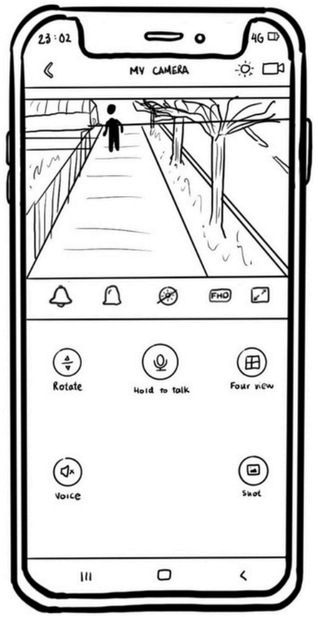
AWear is a clip-on, motion-sensing camera that alerts the user of potential dangers behind them. Its lightweight plastic clip attaches to clothing or personal belongings, such as a belt, shirt hem or backpack, positioning the compact camera to face behind the user.

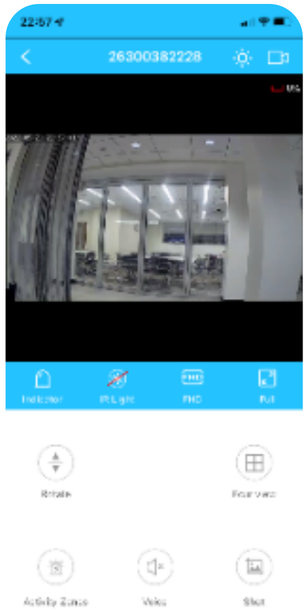




Accompanying App

Wirelessly connected to the camera, the app sends push notifications when motion is detected, including images captured behind the user. Users can view live footage and remotely record, replay or download recordings stored on the camera's SD card.





Instructions for Use



Video Showing How to use Awear

1. Attach the clip to clothing or a personal belonging, positioning the camera to face behind you.
2. Switch on the device and connect it to the accompanying app.
3. Receive notifications when the camera detects movement.
4. Open the app to view captured images or live footage and access recordings.

Clasping Companion

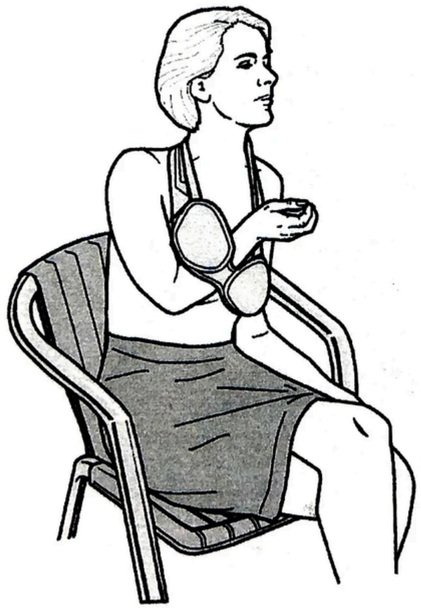
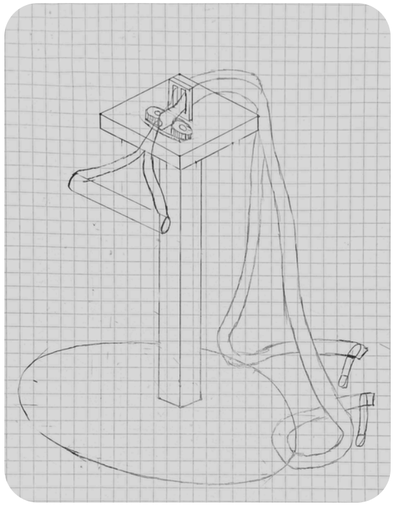


Illustration of the one-handed dressing challenge



Design Thinking & Communication Course | Shirley Ryan Ability Lab

Project Manager | January - April 2022

Understanding User Needs

Conducted online research and interviewed an occupational therapist to understand the dressing challenges and mobility needs of women with hemiplegia.

Key Insights

- Reduced sensation and movement in the affected arm limits its ability to assist with dressing.
- Fastening a bra was identified as one of the most challenging steps when dressing one-handed
- Taking longer than two minutes to put on a bra was reported as a source of frustration.

Design Requirements

Research and user feedback informed four key requirements: :

- One - hand operation
- Compatible with existing bras
- Reusable and durable
- Target cost: ~ $25 - $30

Initial Concept

Explored a strap-based mechanism to bring bra fasteners together and enable one-handed fastening.

Clasping Companion Prototype Design & Testing

Final Prototype

Clasping Companion holds the bra ends in a loop around the user's torso. Pulling the strap tight brings the hooks and eyes together, enabling one -handed fastening.



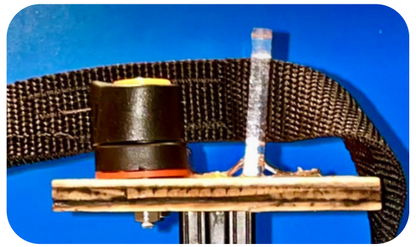
Design Features

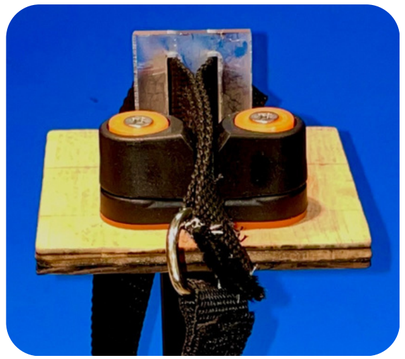
Sailing cleat and belt guide - guides the strap and maintain tension during fastening

Polyester strap and mitten clips - connects the bra ends to the strap and hold them securely during tightening.

Pole and wooden platform - positions the fastening mechanism to help align the bra clasps below the breasts.

Foam handle and stabilising base - provides a comfortable grip for pulling while the base is secured beneath the user's legs.





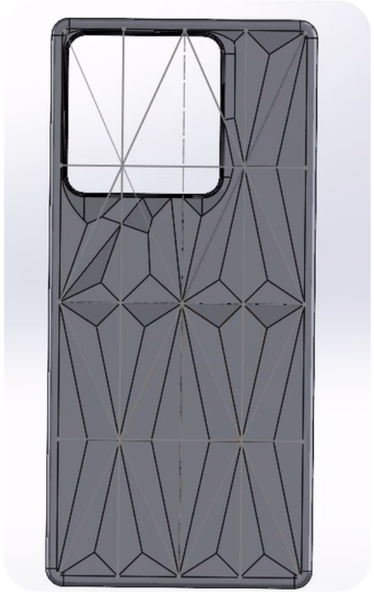


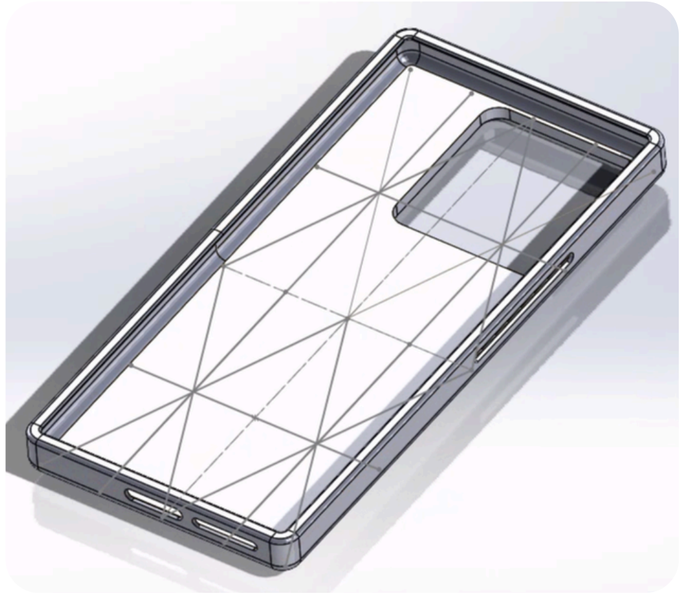
How it Works

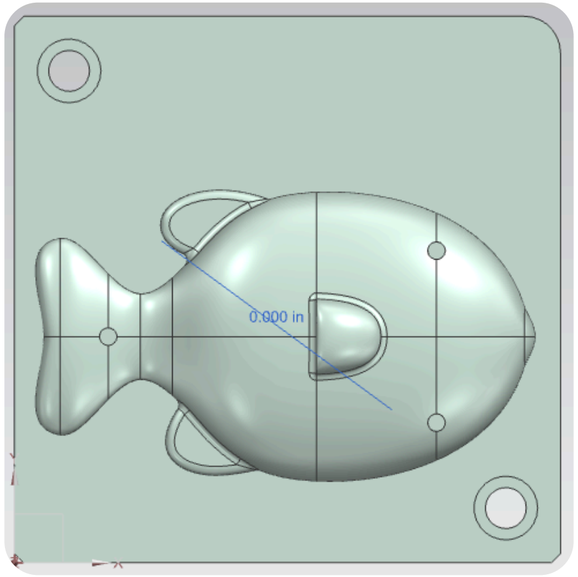
1. **Attach** - connect the bra ends to the strap using the mitten clips.
2. **Position** - place the loop around the torso using the unaffected arm.
3. **Tighten** - pull the foam handle to bring the bra clasps together.
4. **Fasten** - use one hand to clasp the bra, then remove the clips.

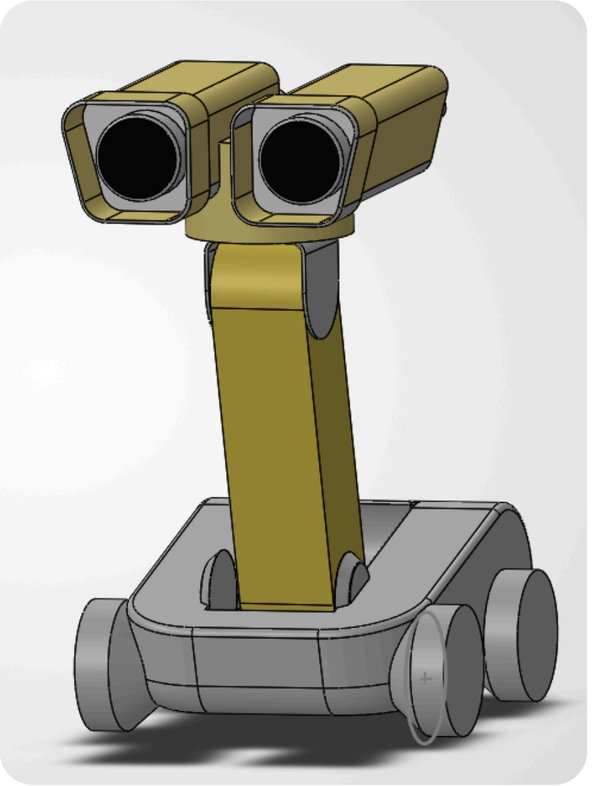


CAD Designs

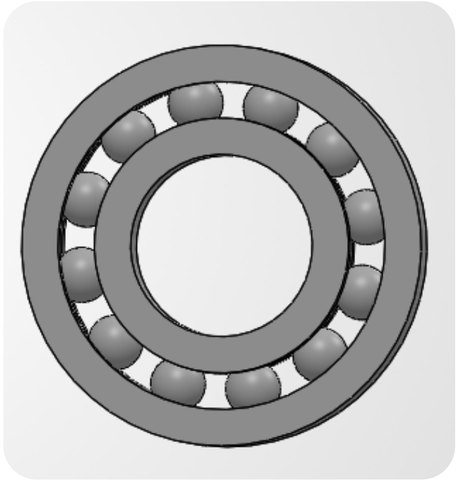


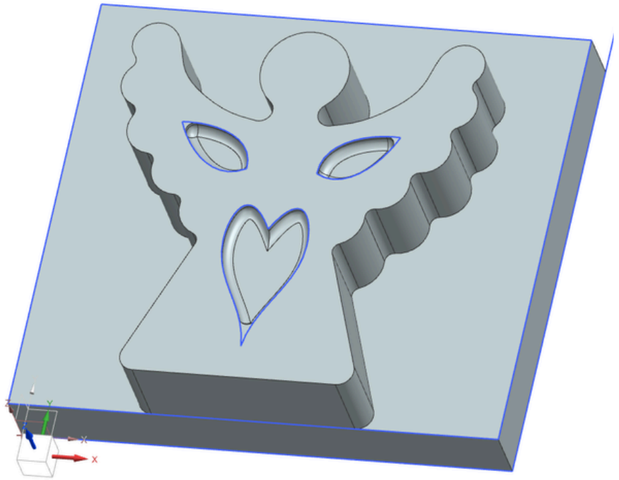




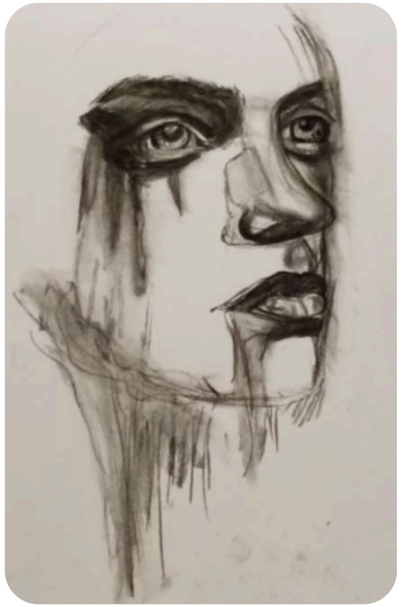




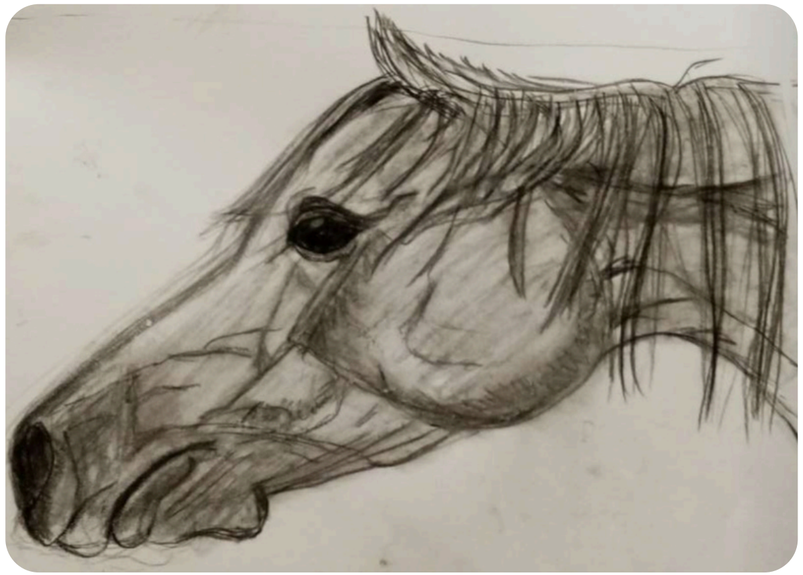


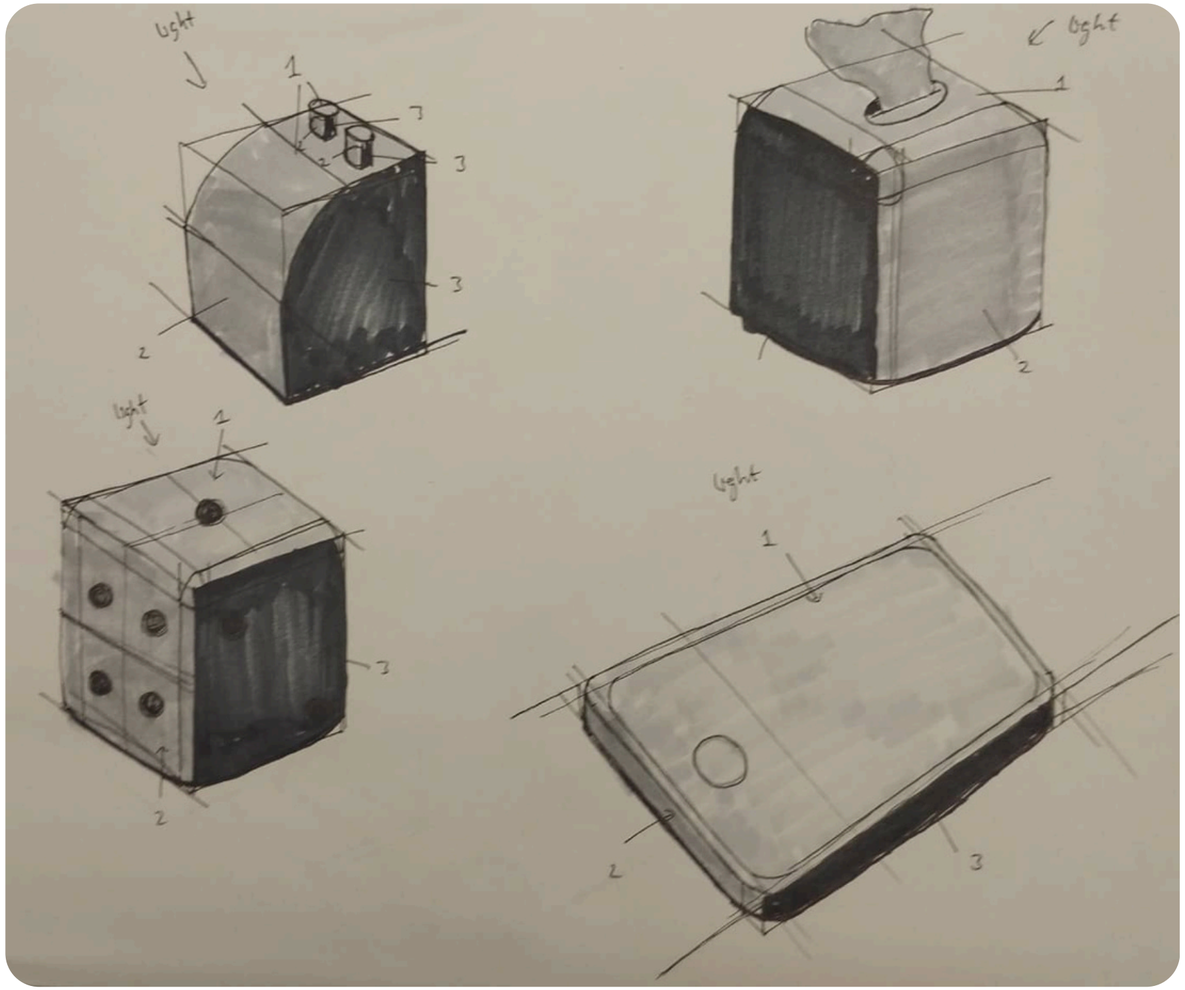


Sketches









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